An expert-performance perspective of research on medical expertise: the study of clinical performance

K Anders Ericsson

CONTEXT Three decades ago Elstein *et al.* published their classic book on medical expertise, in which they described their failure to identify superior performance by peer-nominated diagnosticians using high- and low-fidelity simulations of the everyday practice of doctors.

OBJECTIVE This paper reviews the results of subsequent research, with a particular emphasis on the progress toward Elstein *et al.*'s goal of capturing the essence of superior clinical performance in standardised settings in order to improve clinical practice.

RESULTS Research following publication of Elstein *et al.*'s book was influenced by laboratory research in cognitive psychology, which resulted in a redirection of its original focus on capturing clinical performance in practice to studies of changes in cognitive processes as functions of extended clinical experience. There is currently renewed interest in linking laboratory research with studies of the acquisition of superior (expert) performance in the clinic.

CONCLUSIONS Research on medical expertise and simulation training in technical procedures and diagnosis provide exciting opportunities for establishing translational research on the acquisition of superior (expert) performance in the clinic by capturing it with representative tasks in the laboratory, reproducing it for experimental analysis, and developing training activities, such as deliberate practice, that can induce measurable improvements in performance in the clinic.

KEYWORDS clinical competence/*standards; physicians/*standards; decision making; clinical medicine/*standards.

*Medical Education 2007: **41**: 1124–1130*
doi:10.1111/j.1365-2923.2007.02946.x

Department of Psychology, Florida State University, Tallahassee, Florida, USA

Correspondence: K Anders Ericsson, Department of Psychology, Florida State University, 1107 West Call Street, PO Box 3064301, Tallahassee, Florida 32306-4301, USA. Tel: 00 1 850 644 9860; Fax: 00 1 850 644 7739; E-mail: ericsson@psy.fsu.edu

INTRODUCTION

It is now over 30 years ago since Simon and Chase[1] published their first proposal for a general theory of expertise and almost 3 decades since the classic book by Elstein *et al.*, *Medical Problem Solving: an Analysis of Clinical Reasoning*,[2] appeared. It is timely to review the progress. Last year, the first *Cambridge Handbook of Expertise and Expert Performance*[3] was published, demonstrating that the field of research on expertise has matured into an integrated discipline. Within the last 5 years over 15 books on expertise have been published.[4] Today, research on expertise is taking on particular relevance as federal funding agencies are promoting translational research,[4,5] where scientists have to establish a direct connection between important phenomena in the clinic, such as outcomes of treatments in different hospitals,[6] and laboratory research on the causes of the same phenomena, such as individual differences in the skills of surgeons. Furthermore, medicine is studying methods to measure and pay for performance.[7]

This paper will begin by reviewing Elstein *et al.*'s[2] research and how some of their failures to identify superior clinical performance influenced subsequent investigators and their research agendas. The expert-performance approach will be briefly described and its framework used to put research on medical expertise into perspective.

© Blackwell Publishing Ltd 2007.

Overview

What is already known on this subject

Research on medical expertise has traditionally identified experienced clinicians as experts and then compared their diagnostic processes with those of novices, such as medical students.

What this study adds

This article sketches an alternative approach to medical expertise based on the study and analysis of superior (expert) performance in the clinic and outlines how laboratory research on medical expertise can remain closely connected to superior clinical performance.

Suggestions for further research

Research on medical expertise might establish direct links to robust performance phenomena in the clinic, enabling empirical examination of theoretical assumptions guiding current research. This should also enable measurement of longitudinal development of performance of expert clinicians and describe their associated deliberate practice activities.

THE ORIGINS OF RESEARCH INTO MEDICAL EXPERTISE

Modern research into expertise can be traced back to de Groot's[8,9] classic research on expertise in chess, where he demonstrated the possibility of studying the thought processes of world-class chess players by giving them representative tasks that captured the challenges they would regularly encounter during matches in chess tournaments.

Elstein *et al.*[2] developed a similar research programme to capture superior (expert) performance in diagnosis and treatment. First, Elstein *et al.*[2] carefully designed representative tasks with situations that 'a general internist practicing in a community hospital of moderate size could be reasonably expected to see' with a room 'designed to resemble a physician's office' and actors who 'were carefully trained to simulate the patients'. Second, they searched extensively for expert diagnosticians with superior skills and contacted practising doctors in Michigan for nominations of the 'best diagnosticians known to the respondent', and to identify doctors with the most nominations along with a group of typical doctors. Finally, Elstein *et al.*[2] collected data on doctors' thought processes by videotaping doctor–patient interactions, especially comments made by doctors as if they were 'reviewing cases with students'. In addition, the doctors also viewed their videotapes and reported their memories of thoughts they had had during the videotaped interactions.[2] (pp 47–8)

Based on their analysis of their extensive data, Elstein *et al.*[2] concluded that the objective performance of 'the best diagnosticians' did not differ significantly from that of typical doctors, and nor did cognitive processes differ systematically between the 2 groups. Further, their original conclusions about the general structure of the diagnostic process were later shown to lack rigorous statistical support.[10,11]

Some reactions to Elstein *et al.*'s failure

The failure of Elstein *et al.*'s studies[2] to uncover significant individual differences in diagnostic performance had several consequences. In their book, Elstein *et al.*[2] reviewed possible reasons for the failure, such as problems with recruitment, problems with the tasks and lack of statistical power. In a retrospective assessment 10 years later, Elstein *et al.*[10] suggested that medical doctors would not know enough about their colleagues' diagnostic performance to be able to nominate superior diagnosticians accurately. They concluded that 'years of experience, specialty board certification and (occasionally) academic rank or responsibility have become the indicators of expertise'.[10] (p 22) If medical expertise can be assumed to develop gradually as a function of years of experience, then investigators would not necessarily have to study the highest level of expertise – the specialists – but could study the development toward expertise in more accessible populations, such as medical students and interns. Similarly, the failure to capture individual differences in expert diagnostic performance with either costly high- or low-fidelity simulations made the issue of capturing expertise mute and permitted researchers to explore phenomena related to the acquisition of diagnostic reasoning more freely. In fact, several pioneering researchers adapted experimental paradigms from cognitive psychology for studying prototypes of concepts[12] and exemplar-based categorisation.[13] Patel and Groen[14] adapted the paradigm for studying chess experts' memory for briefly presented chess

positions[15] to the domain of medicine. They had medical students and specialists read a case description of a patient for a couple of minutes, then recall as much as they could in writing and subsequently write out a description of the underlying pathophysiology of the case before reporting their diagnosis. Their focus was thus more on the structure of the information about the patient held in the long-term memory *after* all the relevant information had been processed and the diagnosis generated.[16,17] More generally, these changes in paradigms reduced the connections to superior diagnostic performance in the clinic.

Subsequent research on medical expertise and clinical performance

Research by Bordage[12] comes closest to continued pursuit of Elstein *et al.*'s[2] research agenda, in part because Elstein[18] changed his research focus to statistical decision theory as a more promising area for improving the quality of clinical practice. Bordage[12] continued to use 'think aloud' protocols to study individual differences in diagnostic reasoning, with a new focus on discovering differences in the structure and organisation of knowledge. Bordage developed training interventions to induce the development of elaborated semantic representations of reported symptoms, but there was no improvement in diagnostic accuracy among medical students. He concludes his paper in this issue with a proposal for closer interaction between theoretical laboratory research and training in medical school.

The research agenda of Schmidt and Rikers[19] is directly linked to work by Patel and Groen,[14] who took issue with Elstein *et al.*'s[3] model for the generation and elimination of diagnostic hypotheses in favour of Simon and Chase's[1] theory of expertise, which favoured mechanisms with pattern recognition-based retrieval from memory. Their paper provides a very thoughtful review of the issues regarding forward and backward reasoning, the non-monotonic relationship between immediate memory and expertise, and the role of biomedical knowledge in supporting expert diagnostic reasoning. With respect to relevance to clinical practice, several of the studies conducted by their group have moved closer to analyses of concurrent thought processes on tasks representative of diagnosis in the clinic. Whereas Patel and Groen's paradigm[14] focused on the immediate recall of descriptions of patients and written posthoc explanations, studies carried out by their group had their participants think aloud during their reading of patient descriptions, with the main task of generating a diagnosis rather than immediate recall. A further step in this development is provided by Monajemi *et al.*[20], who studied diagnosis in the context of the goal of treating patients, thus reinstating the task primarily studied by Elstein *et al.*[2] Pertinent to contextual effects, Schmidt and Riker[19] discuss the possibility of differences in experts' cognitive processes during diagnosis in actual clinical practice and in experimental situations and propose further research on differences between the 2 contexts.

The research group at McMaster University, led by Norman, is advocating a research programme that is most clearly opposed to Elstein *et al.*'s[2] central methodological assumptions. Norman *et al.*[13] propose that 'expert and novice problem solving is based on, to some degree, similarity to a prior specific exemplar in memory'[13] as a result of non-analytic reasoning (NAR). Given that NAR 'is presumed to be inaccessible to introspection',[13] this group does not collect verbal reports of thinking and relies primarily on laboratory experiments with experimental paradigms from cognitive psychology. The core paradigm demonstrates experimentally that characteristics of exemplars of a diagnosis (pictures of skin conditions, electrocardiograms [ECGs]) presented during an initial study and practice phase (with a typical duration of around an hour) can bias subsequent diagnoses of other cases (presented with similar stimuli). For example, Young *et al.*[21] taught undergraduate psychology students 4 different pseudo-psychiatric diagnoses and found a bias induced by the studied case examples when subjects were tested immediately after the study and practice phase. Drawing on results with this general paradigm, Norman *et al.*[13] report significant biases for psychology students, medical students and professionals below the level of specialists. Norman *et al.*[13] acknowledge that specialists may be different and may rely on sequential reasoning to diagnose challenging cases. In an interesting study Eva *et al.*[20] demonstrate that it may be possible to reduce these biases in psychology students by inducing both analytic and non-analytic reasoning. However, Eva *et al.*[22] caution against generalisation of their results, and raise some very relevant issues about the generalisability of results from short-term training studies with novices to the acquisition of diagnostic performance that normally takes years of study and internship to acquire in clinical settings.

In their review of the current state of research on medical expertise, Mylopolous and Regehr[23] argue that researchers have focused too much on the effects of extensive experience in diagnostic performance.

According to these authors, contemporary researchers assume that 'most novices eventually become experts. It is presumed that with more experience automatically comes the accrual of a greater (or better) resource base on which to rely, suggesting that expertise is an automatic and inevitable consequence of experience'.[23] By contrast, Mylopolous and Regehr[23] propose that this assumption is only true for some experts, who are skilled in executing routine procedures (routine experts), but that there are other experts (adaptive experts), who 'continue to grow only because of their intentional engagement in ''progressive problem solving''; the continual re-investment of cognitive resources into creating not merely better performance, but in fact better understanding [of] the problem of their domain'.[23] They conclude that current medical education trains routine experts and that new directions are necessary to train adaptive experts.

All the reviews in this special issue[12,13,19,23] acknowledge directly or indirectly that research into medical expertise during the past 3 decades has not identified medical professionals with reproducibly superior performance in their daily clinical practice and thus has not captured the associated phenomena in laboratories, as Elstein *et al.*[2] set out to do. Most researchers of medical expertise seem to have been influenced by early general theories of expertise,[1] with their focus on the differences between the cognitive processes of medical students (novices) and those of experienced clinicians (experts).[23]

Toward the study of superior (expert) performance in clinical practice

The distinction between experienced clinicians and clinicians with superior performance has received hardly any attention among researchers of medical expertise, except for Mylopolous and Regehr's[23] review. In the broader field of expertise research this important distinction has a long history. In the early 1990s my collaborators and I started to criticise a definition of expertise based on years of experience and discussed evidence that the amount of domain-related experience is only weakly related to objective performance of representative tasks for many types of expert, such as chess players, amateur athletes, auditors, stockbrokers and other types of professionals. More recent reviews and meta-analyses of thousands of experienced health professionals show weak or non-existent correlations between performance on representative tasks and years of professional experience after the completion of education.[24–26] In fact, for many types of performance there is a negative correlation with years of experience, which suggests a decay in previously acquired skills.[24–26]

The expert-performance approach

Ericsson and Smith[9] proposed an alternative approach to the study of expertise (now referred to as the expert-performance approach). This approach is based on identifying individuals (expert performers) who exhibit superior performance on tasks that capture the essence of expertise in the critical domain. In the case of medicine, this would correspond to surgeons and clinicians who produce better treatment outcomes, and more accurate diagnoses for adverse conditions that can be treated effectively with established therapies. Recent reviews[24,27] show that studies of expert performers with objectively superior performance in clinical medical settings exist, but are rare.

Reproducing the superior (expert) performance in the laboratory

The expert performance approach starts with an analysis of naturally occurring behaviour,[8,9] such as matches between chess masters and rapid diagnoses in the emergency room. It then identifies key situations, where a critical immediate action, such as a chess move or a corrective action during a surgical problem, is demanded, and the appropriateness of the performed actions can be determined after the fact. These situations are then presented as videos, pictures or simulated environments to other individuals who differ in skill levels and who are instructed to generate their best immediate action. By presenting a sequence of these representative situations it has been possible to capture objective performance in different domains, such as chess, music, individual sports and scrabble, which is closely related to performance in tournaments and competitions. It is possible to measure these abilities using selection of the best move. For example, with 30 different selection tasks (10–20 minutes of testing), it is possible to attain a validity that approaches measures based on the outcomes of matches that correspond to 50–200 hours of tournament play.[4,28]

Since Elstein *et al.* performed their studies,[2] tasks in medical diagnosis have, nearly always, involved typical and routine cases. By contrast, demonstrations of superior diagnostic performance are nearly always associated with challenging cases,[13,25,26,29] which reflect rare, if ever, directly experienced conditions, such as infrequent diseases or unusual combinations

of disease states. These are the types of cases that require 'adaptive expertise'[23] and are associated with rich protocols of pathophysiological reasoning.[13]

Analysis of the mechanisms that mediate superior performance

The establishment of reproducible superiority for representative tasks that capture performance in the clinic permits researchers to analyse the mechanisms responsible for this performance by tracing performance with latencies, eye movements and concurrent or retrospective reports.[30] Research on expertise has illustrated how 'think aloud' protocols during superior representative performance can identify complex skilled mechanisms that can later be validated by specially designed experimental manipulations.[30]

Once the mechanisms mediating the superior performance have been identified, the third step in the expert performance approach involves studying how these mechanisms developed or were acquired. Superior performance on the most challenging tasks is linked to planning, complex reasoning, self-monitoring and evaluation, which require representations and acquired skills to support increased demands for working memory in the form of long-term working memory.[16,17,31] The challenge is to explain how these complex mechanisms were acquired by appropriate practice activities.

The acquisition of superior (expert) performance with deliberate practice

A fundamental prerequisite for improvement of performance accuracy through practice is the availability of valid immediate feedback. In most medical activities it often takes weeks or months until a patient's development of his or her medical condition confirms or rejects the validity of diagnoses and associated treatments. Many interns and residents have to learn to diagnose X-rays, such as mammograms, without receiving accurate feedback and must rely on feedback in the form of their teachers' diagnoses of the same images – which is estimated to be around 70% accurate.[25,32] In some activities, such as surgery and emergency medicine, there is often immediate feedback about mistakes and failures, but these types of everyday activities are less appropriate for training and practice.

In numerous other domains it has been possible to identify special practice activities (deliberate practice) that performers' teachers or the performers themselves design to provide opportunities to improve particular aspects of their performance in an environment that allows gradual refinement after problem solving and repeated variations with immediate feedback.[33] In medicine some of the most exciting research has concerned high-fidelity training with simulators, in particular that involving laparoscopic techniques. A recent meta-analysis[34] revealed that enhanced performance resulting from simulator training depended directly on whether the training procedure incorporated the characteristics of deliberate practice,[25,34] such as goal-directed training, with opportunities for repetition with feedback. A subsequent review was able to demonstrate that increases in the amount of practice (meeting the criteria for deliberate practice) were associated with increases in performance.[35] Furthermore, skilled surgeons with excellent performance records have been tested using simulators and their levels of simulator performance were reliably superior to those of untrained interns. The expert performers' achievement has been used to benchmark the speed and control[36] that surgeons in training need to attain before completing their simulator training. This research on simulator training has even demonstrated significant transfer of skills learned in training to actual surgical and clinical practice.[37]

Conclusions

Emerging technological innovations in imaging, biochemical analysis and advanced treatment modalities are generating some very exciting opportunities for research on medical expertise. The development of simulation centres dedicated to training will provide further opportunities for research on training interventions that enhance performance in the clinic, as well as on the measurement and analysis of superior performance on high-fidelity tasks. It should be possible to return to the original vision of Elstein *et al.*,[2] where studies of superior clinical performance lead to improvement of the quality of clinical practice, but one would have to use a different methodological approach, namely, the expert-performance approach. If current research on medical expertise, discussed earlier in this article, were to establish direct links to robust performance phenomena in the clinic, it would be possible to examine empirically many of the theoretical assumptions that guide current research. For example, are the characteristics of the diagnostic learning of psychology and medical students closely related to those of select interns, residents and specialists, who gradually improve their diagnostic accuracy throughout their careers? It should be possible to measure the longitudinal development of

the representative performance of future superior clinical performers and to describe their associated deliberate practice activities. The future of research on medical expertise looks very bright; I expect that this is where the most important discoveries about the optimal development of professional expertise will be made.

Acknowledgements: the author would like to thank Len Hill for comments on an earlier version of this review.
Funding: this study was supported by Florida State College for Women and Edward Conradi Eminent Scholar Endowment from the Florida State Foundation.
Conflicts of interest: none.
Ethical approval: not applicable.

REFERENCES

1 Simon HA, Chase WG. Skill in chess. *Am Sci* 1973;**61**:394–403.
2 Elstein AS, Shulman LS, Sprafka SA. *Medical Problem Solving: an Analysis of Clinical Reasoning*. Cambridge, MA: Harvard University Press 1978.
3 Ericsson KA, Charness N, Feltovich P, Hoffman RR. *Cambridge Handbook of Expertise and Expert Performance*. Cambridge: Cambridge University Press 2006.
4 Ericsson KA, Williams AM. Capturing naturally occurring superior performance in the laboratory: translational research on expert performance. *J Exp Psychol Appl* 2007;**13**:115–123.
5 Zerhouni EA. Translational and clinical science – time for a new vision. *N Engl J Med* 2005;**353** (15):1621–23.
6 Hollenbeck BK, Dunn RL, Miller DC, Daignault S, Taub DA, Wei JT. Volume-based referral for cancer surgery: informing the debate. *J Clin Oncol* 2007;**25** (1):91–6.
7 Rosenthal MB, Dudley DA. Pay-for-performance: will the latest payment trend improve care? *JAMA* 2007;**297** (7):740–4.
8 de Groot A. *Thought and Choice in Chess* [Original work published 1946.] The Hague: Mouton 1978.
9 Ericsson KA, Smith J. Prospects and limits in the empirical study of expertise: an introduction. In: Ericsson KA, Smith J, eds. *Toward a General Theory of Expertise: Prospects and Limits*. Cambridge: Cambridge University Press 1991;1–38.
10 Elstein AS, Shulman LS, Sprafka SA. Medical problem solving: a 10-year retrospective. *Eval Health Prof* 1990;**13**:5–36.
11 McGaghie WC. Medical problem-solving: a re-analysis. *J Med Educ* 1980;**55**:912–21.
12 Bordage G. Prototypes and semantic qualifiers: from past to present. *Med Educ* 2007;**41**:1193–1201.
13 Norman G, Young M, Brooks L. Non-analytical models of clinical reasoning: the role of experience. *Med Educ* 2007;**41**:1140–45.
14 Patel VL, Groen GJ. Knowledge-based solution strategies in medical reasoning. *Cognit Sci* 1989;**10**:91–116.
15 Chase WG, Simon HA. Perception in chess. *Cognit Psychol* 1973;**4**:55–81.
16 Ericsson KA, Patel V, Kintsch W. How experts' adaptations to representative task demands account for the expertise effect in memory recall: comment on Vicente and Wang (1998). *Psychol Rev* 2000;**107** (3):578–92.
17 Patel VL, Arocha JF, Kaufmann DR. Diagnostic reasoning and medical expertise. *Psychol Learn Motiv* 1994;**30**:187–251.
18 Elstein AS. Clinical problem solving and decision psychology: comment on 'the epistemology of clinical reasoning'. *Acad Med* 2000;**10** (Suppl):134–6.
19 Schmidt HG, Rikers RMJP. How expertise develops in medicine: knowledge encapsulation and illness script formation. *Med Educ* 2007;**41**:1133–39.
20 Monajemi A Rikers RMJP Schmidt HG Clinical case processing: a diagnostic versus a management focus. *Med Educ* 2007;**41**:1166–72.
21 Young M, Brooks L, Norman G. Found in translation: the impact of familial symptom descriptions on diagnosis in novices. *Med Educ*.
22 Eva KW, Hatala R, LeBlanc VR, Brooks LR. Teaching from the clinical reasoning literature: combined reasoning strategies help novice diagnosticians overcome misleading information. *Med Educ* 2007;**41**: 1152–58.
23 Mylopolous R, Regehr G. Cognitive metaphors of expertise and knowledge: prospects and limitations for medical education. *Med Educ* 2007;**41**:1159–65.
24 Choudhry NK, Fletcher RH, Soumerai SB. Systematic review: the relationship between clinical experience and quality of health care. *Ann Intern Med* 2005;**142** (4): 260–73.
25 Ericsson KA. Deliberate practice and the acquisition and maintenance of expert performance in medicine and related domains. *Acad Med* 2004;**10** (Suppl):70–81.
26 Ericsson KA, Whyte J, Ward P. Expert performance in nursing: reviewing research on expertise in nursing within the framework of the expert-performance approach. *ANS Adv Nurs Sci* 2007;**30** (1):58–71.
27 Ericsson KA, Lehmann AC. Expert and exceptional performance: evidence on maximal adaptations on task constraints. *Annu Rev Psychol* 1996;**47**:273–305.
28 van der Maas HLJ, Wagenmakers EJ. A psychometric analysis of chess expertise. *Am J Psychol* 2005;**118**:29–60.
29 Norman GR, Trott AD, Brooks LR, Smith EKM. Cognitive differences in clinical reasoning related to postgraduate training. *Teach Learn Med* 1994;**6**:114–20.
30 Ericsson KA. Protocol analysis and expert thought: concurrent verbalisations of thinking during experts' performance on representative task. In: Ericsson KA, Charness N, Feltovich P, Hoffman RR, eds. *Cambridge Handbook of Expertise and Expert Performance*. Cambridge: Cambridge University Press 2006;223–42.
31 Ericsson KA, Kintsch W. Longterm working memory. *Psychol Rev* 1995;**102** (2):211–45.

32 Nodine CF, Kundel HL, Mello-Thoms C, Weinstein SP, Orel SG, Sullivan DC, Conant EF. How experience and training influence mammography expertise. *Acad Radiol* 1999;**6**:575–85.
33 Ericsson KA. The influence of experience and deliberate practice on the development of superior expert performance. In: Ericsson KA, Charness N, Feltovich P, Hoffman RR, eds. *Cambridge Handbook of Expertise and Expert Performance*. Cambridge: Cambridge University Press 2006;685–706.
34 Issenberg SB, McGaghie WC, Petrussa ER, Lee Gordon D, Scalese R. Features and uses of high-fidelity medical simulations that lead to effective learning: a BEME systematic review. *Med Teach* 2005;**27**:10–28.
35 McGaghie WC, Issenberg SB, Petrussa ER, Scalese RJ. Effect of practice on standardised learning outcomes in simulation-based medical education. *Med Educ* 2006;**40**:792–97.
36 van Sickle KR, Ritter EM, McClusky DA III, Lederman A, Baghai M, Gallagher AG, Smith CD. Attempted establishment of proficiency levels for laparoscopic performance on a national scale using simulation: the results from the 2004 SAGES Minimally Invasive Surgical Trainer-Virtual Reality (MIST-VR) learning centre study. *Surg Endosc* 2007;**21** (1):5–10.
37 Park J, MacRae H, Musselman LJ, Rossos P, Hamstra SJ, Wolam S, Reznick RK. Randomised controlled trial of virtual reality simulator training: transfer to live patients. *Am J Surg* 2007;**194** (2):205–11.

Received 24 September 2007; accepted for publication 10 October 2007